

SC4063 NETWORK FORENSIC FINAL PROJECT – ADDENDUM TO PART 1

INCIDENT OVERVIEW

The company is **Apex Global Logistics**, a mid-sized freight coordinator with a global footprint. 48 hours ago, their operations ground to a halt. Their backup servers are encrypted, and they have received a ransom note from the **Lynx** group. The client's IT team panicked and pulled the power on the infected hosts (destroying most volatile RAM artifacts). However, their network engineering team had a specific tap on their core switch for performance monitoring. They have provided us with a **9-day Packet Capture (PCAP)** covering the window leading up to the encryption, as part of Phase 1 incident response.

Download the PCAP from [networkforensic-groupproject.zip](#)

INVESTIGATION DIRECTIVES

The client's C-suite demands answers before they even consider paying the ransom. Based on the preliminary threat intelligence regarding the **Lynx** group, your investigation must answer the following questions in your report:

A. Initial Access (The "Patient Zero")

- We see no evidence of a phishing email on the mail gateway.
- *Hint:* The Lynx group is notorious for purchasing compromised credentials or exploiting public-facing services. Filter your traffic for remote management protocols (RDP, VPN) entering the network from external IP addresses. Look for a successful login followed by an immediate change in traffic behavior.

B. Lateral Movement & Discovery

- The attacker moved from the entry point to the Domain Controller very quickly.
- *Hint:* Look for "noisy" network scanning tools. We suspect they used tools like **SoftPerfect Network Scanner** or **NetExec** to map the environment. You should see rapid connection attempts to port 445 (SMB) or 135 (RPC) across the subnet.
- *Hint:* Did they create new users? Look for DCERPC traffic that corresponds to account creation or group modification.

C. Exfiltration (The "Double Extortion")

- Lynx threatens to release data if not paid. We need proof of what left the building.
- *Hint:* We detected a massive outbound spike to a file-sharing service. The Lynx group has been known to use **temp.sh** for exfiltration.
- *Hint:* Look for large data transfers (HTTP POST requests) and check if the data was compressed (e.g., 7-Zip magic bytes) before leaving.

D. The Payload

- How was the ransomware deployed?
- *Hint:* If they control the DC, did they use RDP to manually drop the file on the servers?